

Erdős Problem #704

EP-704

L1 - TRACTABLE

PARTIALLY SOLVED

Graph Theory

ORIGINAL PROBLEM STATEMENT

Let G_n be the unit distance graph in $\mathbb{R}^n$, with two vertices joined by an edge if and only if the distance between them is 1.

Estimate the chromatic number $\chi(G_n)$. Does it grow exponentially in n ? Does $\lim_{n \rightarrow \infty} \chi(G_n)^{1/n}$ exist?

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>